Threat Model and Security Control Mapping

Privacy-preserving federated learning for healthcare

01 Trust & adversary	ATTACK SOURCE	02 Threat	RISK SCENARIO	03 Security control	PRIMARY MITIGATION
HC Honest-but-curious aggregation server		T1 Gradient reconstruction		M1 CKKS encryption + aggregate-only threshold decryption	
		T2 Membership and property inference		M2 Gradient clipping + DP accountant	
		T9 Blockchain metadata leakage		M9 Permissioned ledger, pseudonymous IDs and hash-only on-chain records	
MC Malicious or compromised hospital client		T3 Model poisoning and backdoor		M3 Bound proof, reputation weighting, aggregate validation and rollback	
		T4 Sybil and reputation manipulation		M4 Consortium CA, unique identities and admission control	
		T7 Client dropout and availability failure		M7 Quorum, timeout and dropout-tolerant threshold	
NET External network attacker		T5 Tampering, MITM and replay		M5 Mutual TLS, signatures, nonce, round ID and timestamp	
		T8 DoS and resource exhaustion		M8 Rate limiting, quotas, circuit breaker and monitoring	
COL Colluding participants		T6 Key compromise or collusion		M6 t-of-n key shares, HSM, rotation and revocation	
BC Faulty administrator or smart-contract vulnerability		T10 Smart-contract logic failure		M10 Contract audit, multisignature upgrade and emergency pause	
		T12 Consent violation or revoked-data use		M12 Consent registry, purpose binding, revocation and retention enforcement	
USER Malicious inference user		T11 Inference extraction and abuse		M11 Authentication, query limits, minimal output and inference audit	

Reading direction: adversary or failure source > threat scenario > primary mitigating control

Threats are grouped by source to improve traceability and eliminate crossing paths.